

The Quantized Mediator: Tadpole-Freedom from the Grading, First-Order Flavor Amplitudes, and the Exchange Process

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Abstract. The graded-net paper realized the Klein selection rules in background-mediated scattering; this note promotes the background to a quantized mediator and states its honest scope at the outset: the existence of the interacting theory beyond perturbation theory is constructive quantum field theory's open frontier and is not claimed here. What is rigorously delivered: the quantized trilinear vertex with a structural theorem, its exact first-order amplitudes, and the tree-level exchange — the program's first genuinely interacting process. **Theorem (tadpole-freedom from the grading):** every Klein-conserving triple $(A, B, A \oplus B)$ has three *distinct* nonzero components (verified 210/210), so all three legs of the vertex $V = \lambda \int \mu g_{CAB} j^C j^A \partial j^B$ land on independent commuting currents: **the vertex admits no self-contractions** — no normal-ordering ambiguity, no vacuum diagrams at $O(\lambda)$, no self-energy from a vertex's own legs at leading order. The grading is an automatic normal ordering. The first-order amplitudes take the closed form $M = \lambda g_{CAB} \int \mu \varphi_C \varphi_A (\partial \varphi_B)^*$, and the verified pattern is exact: the τ -cycle amplitudes are identical to all computed digits (614.748727 for each of $1 + \text{med}(3) \rightarrow 2$, $2 + \text{med}(1) \rightarrow 3$, $3 + \text{med}(2) \rightarrow 1$, with the vertex sign $+1$ in every cycle image — the tensor's τ -invariance operating at the quantum amplitude level); the Klein-forbidden channel is identically zero; and crossing distinguishes what no classical background could — absorption (614.7) and emission (120.1) carry distinct kinematics on the same vertex. At $O(\lambda^2)$, the exchange process $1+1 \rightarrow 2+2$ through a grade-3 mediator is computed at tree level: finite, Klein-composing (each vertex conserves the charge; the total ledger closes $0 \rightarrow 0$), and τ -cycle invariant through $g^2 = 1$. Beyond first order, the perturbative-algebraic framework is the named scaffold, with tadpole-freedom thinning the divergence structure it must manage; the coupling's value and the flavor hierarchies remain, as everywhere in this series, the state stratum's to determine.

1. The frontier, scoped honestly

Promoting the mediator from classical background to quantum field crosses from exactly solvable models into interacting quantum field theory, whose nonperturbative existence in this setting is an open problem of constructive QFT — stated plainly, claimed by no result below. The rigorous territory is exact perturbative structure: the vertex as a well-defined operator, first-order amplitudes in closed form, tree-level exchange, with the perturbative-algebraic framework (compatible by design with the measurement machinery already installed on this net) as the scaffold for higher orders.

2. Theorem: tadpole-freedom from the grading

Theorem. In every Klein-conserving triple $(A, B, C=A \oplus B)$ with $A \neq B$ nonzero, the three indices are pairwise distinct: $C = A$ would force $B = 0$ and $C = 0$ would force $A = B$. Hence every leg of the quantized vertex

$$V = \lambda \int \mu(u) g_{CAB} j^C(u) j^A(u) \partial j^B(u) du$$

acts on a different, commuting current component. There are no self-contractions: the vertex needs no normal ordering, generates no vacuum diagrams at first order, and produces no self-energy from its own legs at leading order. Verified over all 210 triples. ■

The Klein grading, adopted for selection rules, turns out to also be an ultraviolet gift: the algebra's charge bookkeeping automatically removes the diagrams whose subtraction is elsewhere the first act of renormalization.

3. First-order amplitudes: the rules survive quantization

With one-quantum wavepackets and their positive-frequency profiles $\varphi_X(u)$, the absorption amplitude is closed-form:

$$M(A + \text{med}(C) \rightarrow B) = \lambda g_{CAB} \int \mu(u) \varphi_C(u) \varphi_A(u) (\partial \varphi_B)^*(u) du.$$

Verified, with equalities and zeros the claims (magnitudes are convention-bound): the τ -cycle amplitudes are *identical to all digits* — 614.748727 for all three cycle images, the vertex sign $+1$ in each, the tensor's τ -invariance now a statement about quantum transition amplitudes; the Klein-forbidden $1 + \text{med}(3) \rightarrow 3$ vanishes identically through $g = 0$; and crossing appears where the classical background could not show it — emission, $A \rightarrow B + \text{med}(C)$, carries the conjugated mediator profile and distinct kinematics (120.149 against absorption's 614.749) on one and the same vertex. Jordan diagonality extends to the mediator: as an observable it is generation-diagonal; as dynamics it is exactly the generation-changer the antisymmetric channel prescribes.

4. The exchange process

At $O(\lambda^2)$, tree level, the program's first interacting amplitude: $1 + 1 \rightarrow 2 + 2$ by grade-3 mediator exchange, $M_{\text{exch}} = \lambda^2 \iint \mu(u) \varphi_1(\partial \varphi_2)^* \mu(v) \varphi_1'(\partial \varphi_2')^* W_{\text{med}}(u-v) du dv$ with W_{med} the corpus's positive-frequency current kernel carrying its $i\epsilon$. Computed: finite, with the Klein ledger closing exactly — each vertex conserves the charge, the process totals $0 \rightarrow 0$ — and τ -cycle invariant since $g^2 = 1$ in every cycle image. Two generation-one quanta scatter into generation two by exchanging the third generation's charge: the selection rule, now doing dynamics.

5. Beyond first order

Three statements frame the road. The perturbative-algebraic framework is the rigorous scaffold for higher orders on this net, and tadpole-freedom thins the divergence structure it must manage from the start. The sector bookkeeping of trilinear vertices — whether the derivative leg keeps loop-level processes memory-inert as it does at tree level — is flagged as the next analysis, not assumed. And the values — λ , the mediator profile, the hierarchies the democratic kinematics cannot supply — remain the state stratum's to determine: the

algebra has now fixed everything it can fix, exactly, through thirty-four papers, and what it leaves free it leaves free identifiably.

6. Scope

Five items. (i) All statements are perturbative; nonperturbative existence is open and stated so. (ii) Amplitude magnitudes depend on packet and normalization conventions; the claims are the equalities, the zeros, and finiteness. (iii) All fields are chiral and massless; mediator dispersion and mass are not modeled. (iv) No Standard-Model identification is claimed. (v) The construction program continues: loop structure under the pAQFT scaffold, the memory ledger at higher orders, and the state-stratum determination of couplings — the deep standing question.

References

- Companion papers: *The Graded Net: Klein-Charged Currents, the Jordan Diagonality of Observables, and the Selection Rule as Scattering*; *The Generation Grading: The Klein Selection Rule, Democratic Gauge Universality, and the Cyclic Generation Symmetry*; *The Measurement Gate: The Probe Scheme on the Net, Unsharp Induced Observables, and Sector-Preserving Scattering*; *The Charged Coupling: The Infrared Catastrophe as a Power Law, Sector Pumping, and the Memory-Charge Relation*.
- Fewster, C. J. & Rejzner, K., "Algebraic quantum field theory — an introduction," arXiv:1904.04051.
- Fewster, C. J. & Verch, R., "Quantum fields and local measurements," *Comm. Math. Phys.* 378 (2020), 851–889.

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